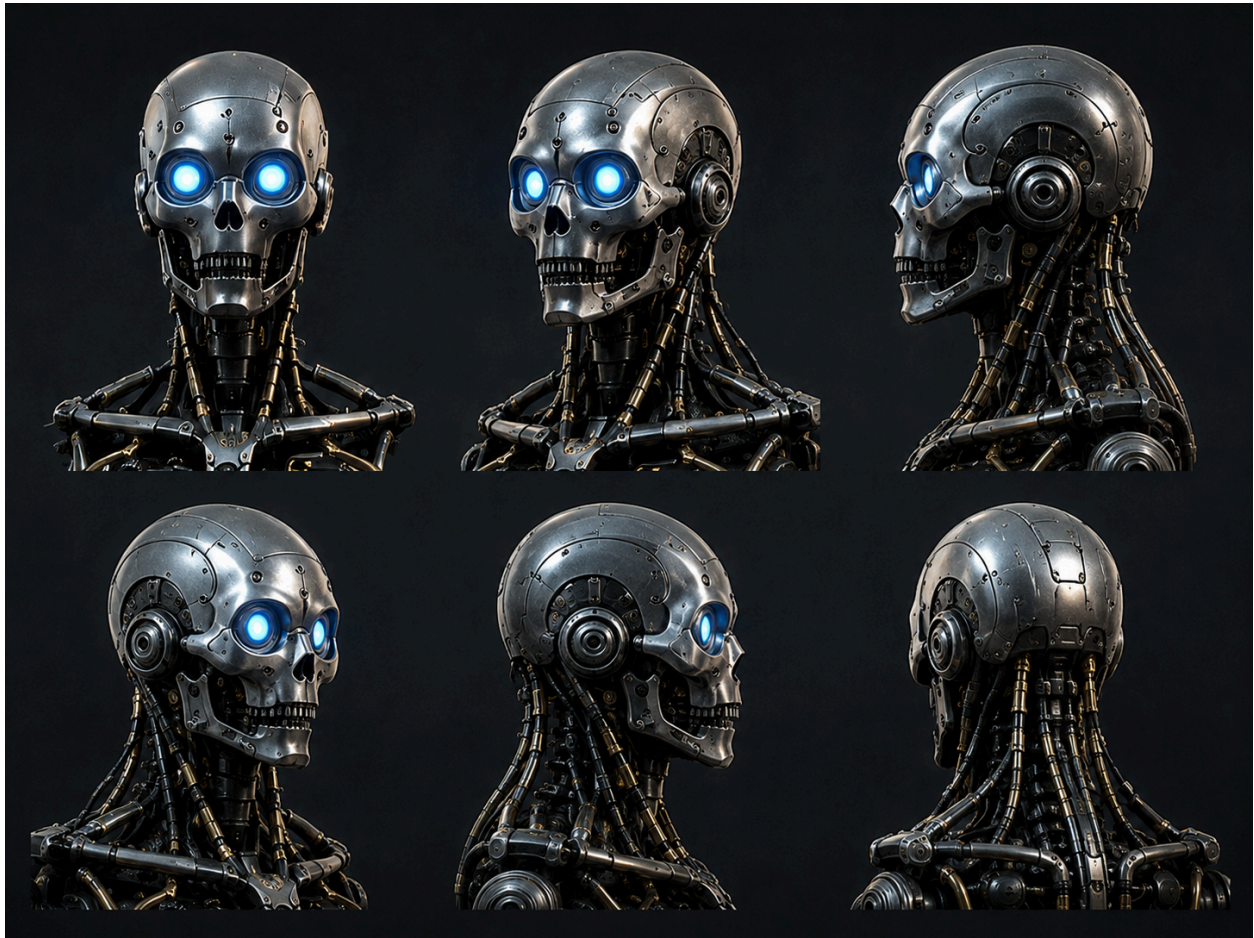


Visual Specification Sheet: Comprehensive documentation of dimensions, control zones, material palettes, and 3D modeling/rigging references.



Robot Head Assembly

1. Overview & Scale Anchor

The full robot design is anchored to a total height reference of 200 cm.

The head component is specified at a fixed height of 26 cm, representing exactly 13% of the total body proportions.

All secondary measurements detailed in this document are visual estimates derived from primary front-facing reference data and must be validated during subsequent multi-angle blockout stages.

2. Control & Sensor Map

The mechanical and sensor topography of the assembly is segmented into seven core operational zones:

1. **Cranial Access Shell:** Removable top service cover detailing explicit panel seams and hardware recesses for M2-M3 fasteners.
2. **Left Optical Pod:** Primary sensor housing containing the camera lens assembly, a blue emissive LED status ring, and an integrated depth/range finder behind a protective convex lens.
3. **Right Optical Pod:** Mirrored stereoscopic imaging module. Rigging requirement: maintain absolute operational alignment with the left eye focal center.
4. **Central Nose Vent:** Multi-purpose structural aperture functioning as an acoustic intake grid or a thermal exhaust vent for internal processing units.
5. **Mandible Assembly:** Servo-driven lower jaw plate integrating an active speaker grille and specialized teeth-like structural heat-sink ribs for heat dissipation.
6. **Temple Side Module:** Dedicated lateral interface accommodating localized audio pickups, short-range proximity sensors, or auxiliary maintenance data transfer ports.
7. **Neck Interface:** Multi-axis yaw/pitch/roll joint stack complete with a structured cable harness routing system, high-load bearing rings, and internal power/data pass-through lines.

3. Dimension Blockout Specification

The following table outlines the spatial boundaries and structural constraints required for initial 3D blockout stages:

Feature	Size	3D Modeling Note
Total head height	26 cm	Fixed absolute value from master body proportion sheets.
Cranial cap	~10 cm	Measurement span from the top outer shell down to the brow line.
Optic / face band	~8 cm	Encompasses the optical pods,

Feature	Size	3D Modeling Note
		central nose bridge, and lateral cheek structures.
Mandible / lower jaw	~8 cm	Includes the teeth cooling grille assembly and the lower chin protective housing.
Maximum front width	~16 cm	Total temple-to-temple horizontal shell width constraint.
Eye lens diameter	~4.0 cm	Outer dimension for each of the visible blue emissive glass lenses.
Eye-center spacing	~9 cm	Interpupillary distance measured from left lens center to right lens center.
Head depth	~18 cm	Estimated front-to-back spatial boundary for initial 3D blockout.
Neck bearing interface	~7 cm dia.	Diameter configuration for the round mounting collar positioned below the jaw line.

4. Material Palette & Finishes

Component	Material Allocation & Specifications
Outer cranium	Dark anodized aluminum or premium titanium alloy; finished in a fine brushed gunmetal treatment with a panel thickness profile of 2–3 mm.
Face frame	Precision CNC-machined aluminum ribs structurally paired with black oxide finished inset plates.
Mandible / teeth	Chemically blackened stainless steel components, reinforced via hardened alignment pins and heat-sink ridges.
Eye lenses	Convex sapphire crystal or impact-resistant polycarbonate; outfitted with internal blue diffusers and high-output emissive LED rings.
Fasteners	Standard M2 and M3 hex head socket screws and heavy-duty rivets featuring a durable matte black oxide coating.
Cable harness	Heavy-duty braided PTFE-insulated high-conductivity copper and fiber-optic data networks, supported by molded rubber strain relief boots.
Neck coupler	Alloy steel mechanical bearing rings reinforced with carbon-fiber composite layups or dense black polymer protective shrouds.

5. 3D Design & Rigging Guidelines

Modeling Structure & Hierarchy

To optimize material application and control animation parameters seamlessly, the mesh topology must be organized as distinct sub-objects.

Ensure the following assets remain separated within the assembly hierarchy: cranium shell, inner structural face frame, individual left and right eye pods, active jaw assembly, lateral temple side modules, neck mechanical coupler, fastening screws, and dynamic cable harness bundles.

Rig Controls & Kinematics

The primary animation rig must feature a parent neck joint supporting independent yaw, pitch, and roll axes.

Each optical pod requires dedicated, independent aim constraints to facilitate realistic tracking motion.

The mandible requires a singular main open/close servo controller mapped with fine-tuned, secondary mechanical offset attributes to simulate lifelike articulation adjustments.

Surface Detail & Hard-Surface Finishing

Incorporate distinct technical detailing to elevate the photorealistic look. Apply precise bevel parameters ranging from 0.8 mm to 2.0 mm on all exposed sharp metal edges.

Model realistic panel gaps, clean screw recess indentations, and subtle edge wear maps targeting high-contact operational corners.

Standard Material Naming Conventions

M_dark_brushed_metal

M_black_oxide

M_eye_blue_emissive

M_eye_glass

M_cable_rubber

M_fastener_dark

M_neck_bearing

Target Animation States

The technical setup must natively accommodate the following standard performance cycles:

- **Idle Scan:** Ambient tracking and environmental reconnaissance pass.
- **Focus Lock:** Direct fixation macro targeting with localized optical adjustment.

- **Diagnostic Blink:** Fast shutter refreshing sequence for environmental sensor recalibration.
- **Jaw Speech Motion:** Articulated mandible movement synchronized with vocal audio tracks.
- **Neck Micro-Stabilization:** High-frequency sub-millimeter motor corrections mimicking balance loops.
- **Service-Mode Eye Dimming:** Energy conservation or system diagnostic standby state.

Critical Proportion Constraint

Important Warning: Apart from the fixed 26 cm total head height, all spatial parameters specified herein are visual approximations derived from a single perspective. It is mandatory to execute a thorough verification and alignment of structural proportions immediately upon generating side-view modeling profiles or point-cloud photogrammetry inputs.